🌐 **Deployment Link:** https://term-4-portofolio.vercel.app/

1. Introduction

During the academic year 2025-2026, I enrolled in the activities coordinated by the Centre for Talent Enrichment (CTE). The primary purpose of the CTE program is to supplement formal undergraduate engineering education with practical, hands-on exposure to emerging technologies and diverse collaborative activities. As a first-year student pursuing a Bachelor of Technology in Computer Science and Engineering (CSE-Core), participating in these specialized workshops allowed me to bridge the gap between abstract programming concepts and physical computing applications.

Engaging in enrichment activities beyond the regular academic curriculum has played an important role in my initial professional development. It provided me with a structured framework to explore fields like robotics, embedded systems, and AI-powered software design. Throughout the year, I participated in four major technical workshops: the Robo Race Line Follower Robot build, the AI RoboDog motion and perception session, the Gesture Tech sensor control lab, and the Base44 Hands-On App Building workshop.

This reflective portfolio documentation serves as an authentic record of my learning journey. It outlines the technical skills, practical workflows, and theoretical concepts I explored. Writing these reflections in a structured format has helped me consolidate my understanding and identify how the topics I learned relate directly to my primary field of computer science. This record stands as a foundation for my future academic projects, technical collaborations, and developmental goals.

2. Overview of Participation

A verified list of workshops attended under the CTE program.

S.No.	Event Name	Date
1	**Robo Race: Line Follower Robot**	28 Apr, 2026
2	**AI RoboDog: Vision & Motion Control**	5 May, 2026
3	**Gesture Tech**	15 Apr, 2026
4	**Base44 Hands-On App Building**	18 Nov, 2025

3. Reflection on Learning Journey (Kolb's Cycle)

Interactive exploration of the Experiential Learning stages.

1. Concrete Experience
2. Reflective Observation
3. Abstract Conceptual

4. Detailed Workshop & Event Learning Records

Select a technical elective from the sidebar below to review its comprehensive syllabus, skills, and academic links.

Robo Race: Line Follower Robot

AI RoboDog: Vision & Motion Control

Gesture Tech

Base44 Hands-On App Building

5. Application of Learning (Transferable Skills Matrix)

A mapping of elective activities to immediate practical applications and future B.Tech engineering uses.

Embedded Control Systems

Robo Race: Line Follower Robot

KEY TECHNICAL CONCEPT

Feedback control loop logic, GPIO pins, and motor speed modulation (PWM).

IMMEDIATE APPLICATION

Debugging physical electronics circuits and basic sensor wiring.

FUTURE B.TECH USE

Implementing motion controllers in microcontrollers and robotics projects.

Kinematics & perception

AI RoboDog: Vision & Motion Control

KEY TECHNICAL CONCEPT

Sensor fusion principles, path planning, and spatial mapping algorithms.

IMMEDIATE APPLICATION

Analyzing autonomous vehicle and navigation systems.

FUTURE B.TECH USE

Designing AI-based pathfinding routines in simulation code.

Signal Processing & HMI

Gesture Tech Workshop

KEY TECHNICAL CONCEPT

Data acquisition from IMUs, noise filtering, and 3D coordinate rotation.

IMMEDIATE APPLICATION

Building head-tracking control or gestural inputs.

FUTURE B.TECH USE

Designing accessibility interfaces and human-computer interactions.

Rapid Software Prototyping

Base44 Hands-On App Building

KEY TECHNICAL CONCEPT

Relational database schema modeling, visual UI layout, and API integration.

IMMEDIATE APPLICATION

Rapid prototyping of web application frontends.

FUTURE B.TECH USE

Structuring full-stack projects and design validation.

6. Self-Assessment Competency Matrix

Honest evaluation metrics across major development dimensions. Preserved from the official submission document.

Competency Scores

Skill Development 4 / 5 (80%)

Application of Learning 4 / 5 (80%)

Confidence & Expression 4 / 5 (80%)

Teamwork & Collaboration 4 / 5 (80%)

Discipline & Commitment 4 / 5 (80%)

Creativity & Innovation 4 / 5 (80%)

Overall Learning Outcome 4 / 5 (80%)

Knowledge of Club Discipline 3 / 5 (60%)

Emotional & Social Growth 3 / 5 (60%)

Interpretation of Results

A review of my self-assessment ratings indicates areas of strength and developmental opportunities. I rated my 'Skill Development', 'Confidence & Expression', and 'Teamwork' at 4 (Good), reflecting the growth I experienced through group problem-solving and technical tasks. Coordinating with peers during the robotics calibration and app building sessions strengthened my collaborative skills.

Conversely, my 'Knowledge of Club Discipline' and 'Emotional & Social Growth' are rated at 3 (Satisfactory). These ratings reflect that as a first-year student, I am still building a fundamental technical vocabulary and adapting to the collaborative dynamics of college-level technical clubs. These scores serve as a realistic benchmark, reminding me to focus on deepening my understanding of core principles in future workshops.

Scale: 1 = Beginner | 2 = Developing | 3 = Satisfactory | 4 = Good | 5 = Excellent

7. Achievements, Recognitions, & Evidence

Verified certificates of completion.

✓ **Verified Achievement**

Base44 Hands-On App Building Workshop Certificate

Figure 7.1: Certificate of Completion - Somula Sudharshan Reddy (Issue Date: 18/11/25)

8. Conclusion & Future Roadmap

Summarizing developmental milestones and future B.Tech engineering project timelines.

Reflection Summary

My participation in the Centre for Talent Enrichment electives during the 2025-2026 academic year has provided a valuable introduction to practical technology. It allowed me to apply the coding and logic concepts learned in my CSE coursework to physical systems and interactive applications. Through these workshops, I gained an initial understanding of feedback control loops, sensor calibration, relational database structures, and HMI mapping.

This experience has shown me that combining programming with hardware or no-code workflows creates highly engaging developer experiences. I have built foundational skills that will serve as a starting point for more complex systems in my later years of study.

Future Project Roadmap

1. **Advanced Embedded Project**

   Build a microcontroller project that incorporates sensor tracking and wireless data logging, applying skills from the Robo Race and Gesture Tech workshops.

2. **Full-Stack Custom Development**

   Study database design (SQL/NoSQL) and backend routing in detail, moving from no-code interfaces to custom code implementations.

3. **Group Collaborations**

   Participate in college hackathons and group robotics events to strengthen teamwork and rapid-prototyping capabilities.

SOMULA SUDHARSHAN REDDY

B.Tech Computer Science and Engineering | Reg: 252U1R1231

Centre for Talent Enrichment (CTE)